



7CS033 - Data Mining & Informatics

**School of Mathematics and Computer Science
Faculty of Science and Engineering**

Coursework by:

Solagbade Akande
2417206
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T6.1 Decision Tree Split Analysis with Information Gain

We are given 6 data points with two binary attributes, X_1 and X_2 , and a class label $Y \in \{1, 2, 3\}$. The goal is to construct a decision tree using a single best split. We evaluate the information gain for both attributes and determine the optimal split.

Dataset Overview

X_1	X_2	Y
1	1	1
1	1	1
1	1	2
1	0	3
0	0	2
0	0	3

Compute Overall Entropy Before Any Split (For the Entire Dataset)

Before any split, we calculate the entropy of the entire dataset to understand how mixed the classes are.

Classes:

- Class 1 ($Y = 1$): 2 data points
- Class 2 ($Y = 2$): 2 data points
- Class 3 ($Y = 3$): 2 data points

Total = 6 points.

The probabilities for each class are:

$$P(Y = 1) = 2 / 6 = 0.33$$

$$P(Y = 2) = 2 / 6 = 0.33$$

$$P(Y = 3) = 2 / 6 = 0.33$$

The entropy is:

$$\text{Entropy} = -\sum P(i) \log_2(P(i))$$

Where: - $P(i)$ is the probability of class i .

$$\text{Entropy}(S) = -(0.33 \log_2 0.33 + 0.33 \log_2 0.33 + 0.33 \log_2 0.33) \approx 1.585$$

Thus, the entropy before any split is **1.585**.

Split by x_1

Let's group the data by X_1 (whether $X_1 = 1$ or $X_1 = 0$):

Group 1: $X_1 = 1$:

The data points are:

$Y = [1, 1, 2, 3]$

Classes:

- Class 1 $\rightarrow 2$
- Class 2 $\rightarrow 1$
- Class 3 $\rightarrow 1$

Count the occurrences of each class for $X_1 = 1$:

- Class 1 appears **twice**
- Class 2 appears **once**
- Class 3 appears **once**
- Total = 4 data points

Calculate probabilities:

- $P(Y = 1) = 2/4 = 0.5$
- $P(Y = 2) = 1/4 = 0.25$
- $P(Y = 3) = 1/4 = 0.25$

The entropy for $X_1 = 1$ is:

$$\text{Entropy}(S_1) = -(0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.25 \log_2 0.25) = 1.5$$

Group 2: $X_1 = 0$:

The data points are:

$Y = [2, 3]$

Classes:

- Class 2 $\rightarrow 1$
- Class 3 $\rightarrow 1$

Count the occurrences of each class for $X_1 = 0$:

- Class 2 appears **once**
- Class 3 appears **once**
- Total = 2 data points

Calculate probabilities:

- $P(Y = 2) = 1/2 = 0.5$
- $P(Y = 3) = 1/2 = 0.5$

The entropy for $X_1 = 0$ is:

$$\text{Entropy}(S_2) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1.0$$

Weighted Entropy After the Split by X_1

The total entropy after the split by X_1 is the weighted average of the entropies of the two groups:

$$\text{Entropy}(X_1) = (4/6 \times 1.5) + (2/6 \times 1.0) = 1.333$$

where:

- $4/6$ is the proportion of samples where $X_1 = 1$ (4 out of 6),
- 1.5 is the entropy of Y for those samples ($Y = [1,1,2,3]$),
- $2/6$ is the proportion where $X_1 = 0$ (2 out of 6),
- 1.0 is the entropy of Y for those samples ($Y = [2,3]$).

Information Gain for X_1

$$\text{IG}(X_1) = E(\text{parent}) - \sum w_i \times E(\text{child}_i) = 1.585 - 1.333 = 0.252$$

Split by X_2

Group 1: $X_2 = 1$:

The data points are:

$$Y = [1, 1, 2]$$

Classes:

- Class 1 $\rightarrow$ 2
- Class 2 $\rightarrow$ 1

The probabilities are:

$$P(Y = 1) = 0.67, P(Y = 2) = 0.33$$

The entropy is $X_2 = 1$:

$$\text{Entropy}(S_1) = -(0.67 \log_2 0.67 + 0.33 \log_2 0.33) \approx 0.92$$

Group 2: $X_2 = 0$:

The data points are:

$Y = [3, 2, 3]$

Classes:

- Class 2 $\rightarrow$ 1
- Class 3 $\rightarrow$ 2

The probabilities are:

$P(Y = 2) = 0.33, P(Y = 3) = 0.67$

The entropy for $X_2 = 0$ is:

Entropy(S_2) = $-(0.33 \log_2 0.33 + 0.67 \log_2 0.67) \approx 0.92$

Weighted Entropy After the Split by X_2

The total entropy after the split is the weighted average of the entropies of the two groups:

Entropy(X_2) = $(3/6 \times 0.92) + (3/6 \times 0.92) = 0.92$

Information Gain for X_2

The information gain is calculated as:

IG(X_2) = $E(\text{parent}) - \sum w_i \times E(\text{child}_i) = 1.585 - 0.92 = \mathbf{0.665}$

Which Split is Better?

- Information Gain for $X_1 = 0.252$
- Information Gain for $X_2 = 0.665$

Clearly, splitting by X_2 is the better choice since it gives a higher information gain.

Breakdown of the Data:***Splitting on X_2 :***

- For $X_2 = 0$ (rows with $X_2 = 0$):
 - $[1,0,3]$ — class 3
 - $[0,0,2]$ — class 2
 - $[0,0,3]$ — class 3

So the class labels for $X_2 = 0$ are:

- 3 (2 occurrences)
- 2 (1 occurrence)

Since **class 3** appears more frequently (2 times), the predicted class label for $X_2 = 0$ would be **3**.

- For $X_2 = 1$ (rows with $X_2 = 1$):
 - [1,1,1] — class 1
 - [1,1,1] — class 1
 - [1,1,2] — class 2

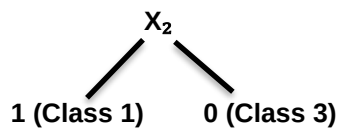
So the class labels for $X_2 = 1$ are:

- 1 (2 occurrences)
- 2 (1 occurrence)

Since **class 1** appears more frequently (2 times), the predicted class label for $X_2 = 1$ would be **1**.

Final Decision Tree

Here's the decision tree based on splitting by X_2 :



Predicted Class in Each Leaf

- If $X_2 = 1$, predict Class 1.
- If $X_2 = 0$, predict Class 3.

Conclusion

- Best split: X_2 , because it gives the highest information gain.
- The decision tree uses X_2 to predict class labels:
 - $X_2 = 1 \rightarrow$ Class 1
 - $X_2 = 0 \rightarrow$ Class 3

T6.2 K-Means Clustering: Calculation

This presents the K-means clustering solution for the given set of 3D points using the initial cluster centers (1,9,14) and (2,18,23).

Dataset

Points:

(1, 9, 14), (2, 18, 23), (3, 30, 30), (4, 21, 9), (5, 9, 17), (6, 25, 32), (7, 36, 25), (8, 10, 12), (9, 38, 45), (10, 1, 2)

Step 1: Choose 2 random initial cluster centers

Initial C1 = (1, 9, 14)

Initial C2 = (2, 18, 23)

Step 2: Compute Euclidean distance and assign points to nearest cluster

Euclidean Distance Formula:

Euclidean Distance between two points p and q:

$$d(p, q) = \sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2]}$$

Step 3: Distance Calculations and Cluster Assignments (Iteration 1)

Point	Dist to C1 (1,9,14)	Dist to C2 (2,18,23)	Assigned to
(1,9,14)	0.00	15.52	C1
(2,18,23)	15.52	0.00	C2
(3,30,30)	28.32	15.81	C2
(4,21,9)	13.75	14.14	C1
(5,9,17)	4.24	14.04	C1
(6,25,32)	23.77	12.25	C2
(7,36,25)	30.79	18.87	C2
(8,10,12)	7.35	16.40	C1
(9,38,45)	41.82	27.29	C2
(10,1,2)	15.52	27.46	C1

Step 4: Recomputed Centroids

New C1: $(1+4+5+8+10) / 5, (9+21+9+10+1) / 5, (14+9+17+12+2) / 5 = (5.6, 10.0, 10.8)$

New C2: $(2+3+6+7+9) / 5, (18+30+25+36+38) / 5, (23+30+32+25+45) / 5 = (5.4, 29.4, 31.0)$

Step 5: Reassign Points (Iteration 2)

Point	Dist to New C1 (5.6,10,10.8)	Dist to New C2 (5.4,29.4,31.0)	Assigned to
(1,9,14)	5.63	30.83	C1
(2,18,23)	14.30	12.57	C2
(3,30,30)	27.52	24.69	C2
(4,21,9)	11.18	24.17	C1
(5,9,17)	6.33	21.15	C1
(6,25,32)	24.22	19.83	C2
(7,36,25)	27.17	19.12	C2
(8,10,12)	3.94	25.67	C1
(9,38,45)	41.79	25.94	C2
(10,1,2)	11.31	38.89	C1

Final Clusters (No Change in Cluster Assignment)

The algorithm converged on the second iteration

Cluster 1: (1,9,14), (4,21,9), (5,9,17), (8,10,12), (10,1,2)

Centroid: (5.6, 10.0, 10.8)

Cluster 2: (2,18,23), (3,30,30), (6,25,32), (7,36,25), (9,38,45)

Centroid: (5.4, 29.4, 31.0)

Final Answer:

Cluster Centers:

- C1: (5.6, 10.0, 10.8)
- C2: (5.4, 29.4, 31.0)

Clusters:

- Cluster 1: (1,9,14), (4,21,9), (5,9,17), (8,10,12), (10,1,2)
- Cluster 2: (2,18,23), (3,30,30), (6,25,32), (7,36,25), (9,38,45)

Iterations: 2 (converged on second)